

STANDARD OPERATING PROCEDURE — PROCESS SAFETY MANAGEMENT

Document No: SOP-PSM-004 | Rev. 5 | Effective: 2023-01-15

Plant Safety Department | Approved: Plant Manager | Review Due: 2025-01-15 | Based on OSHA 29 CFR 1910.119

1. PURPOSE AND SCOPE

This Standard Operating Procedure (SOP) establishes the process safety management requirements for all rotating and pressure-containing equipment in Plant Areas A, B, and C. It implements the requirements of OSHA 29 CFR 1910.119 (Process Safety Management of Highly Hazardous Chemicals) and the facility's internal Process Safety Management (PSM) programme.

This SOP covers: pressure relief device settings, compressor operating limits, pump and valve pressure testing, and management of change procedures for process equipment. All operations and maintenance personnel must be trained on this SOP annually. Training records must be retained for a minimum of 3 years.

2. GENERAL PRESSURE MANAGEMENT REQUIREMENTS

All pressure-containing equipment must be operated within the pressure limits specified in the applicable Mechanical Integrity programme and equipment datasheets. Pressure relief devices (PRDs) must be sized and set in accordance with API 520 and API 521. PRD settings must be reviewed and verified during each planned shutdown or at a maximum interval of 36 months.

The following general principles apply to all pressure systems:

- No PRD shall be set above the Maximum Allowable Working Pressure (MAWP) of the protected vessel or piping.
- PRD set points shall be documented in the Equipment Pressure Relief Register (Form EQ-PRR-001).
- Any deviation from registered PRD settings requires an approved MOC (Management of Change) before implementation.
- PRD testing and certification shall be performed by a certified pressure testing contractor annually.

3. EQUIPMENT-SPECIFIC PRESSURE SETTINGS

3.1 General Compressor Relief Requirements

All compressors in Process Areas A, B, and C must be protected by a dedicated pressure safety valve (PSV) mounted on the high-pressure discharge line. PSV settings must comply with Process Engineering specification PE-2021-047 and must be verified during each annual turnaround.

3.2 Compressor-C1 Relief Valve Settings

Compressor-C1 is protected by PSV-C1-001 (3-inch × 4-inch full-nozzle PSV, Crosby Series HCI, spring code R-14). In accordance with Process Engineering specification PE-2021-047 and the facility's process hazard analysis (PHA) conducted in 2021, the PSV set point for Compressor-C1 is established as follows:

Compressor-C1 pressure relief valve shall be set to 180 psi maximum allowable working pressure (MAWP) as per Process Engineering specification PE-2021-047. This set point accounts for the downstream piping MAWP rating of 200 psi (ANSI Class 150 carbon steel) and provides a 10% margin above the normal maximum discharge pressure of 163 psi observed during high-load operations.

[WARNING] The PSV set point of 180 psi for Compressor-C1 was established based on a 2021 process hazard analysis. Any proposal to change this set point requires a full MOC review with PHA sign-off, not just equipment-level approval. Contact the Process Safety Engineer before adjusting PSV-C1-001.

3.3 Pump Relief Settings

Pump-P3 discharge line is protected by PSV-P3-001 set at 220 psi. All other pump PSVs in Plant Area C are set at 210 psi unless otherwise specified in the Equipment Pressure Relief Register.

4. MANAGEMENT OF CHANGE PROCEDURE

Any change to process safety parameters — including PRD set points, operating pressure limits, equipment isolation schemes, or safety interlock settings — must follow the facility's Management of Change (MOC) procedure (SOP-MOC-001) before implementation. MOC review must include: process hazard analysis update, P&ID; mark-up, operations and maintenance notification, and training completion verification.

Unauthorised changes to PSV settings or operating pressure limits are a **SERIOUS SAFETY VIOLATION** and will result in immediate investigation under the facility's discipline and incident reporting procedures.

5. RECORD KEEPING

- Equipment Pressure Relief Register (EQ-PRR-001): maintained by Mechanical Integrity team.
- PSV test certificates: retained for a minimum of the life of the equipment plus 5 years.
- MOC records: retained for a minimum of 10 years.
- Operator training records: retained for a minimum of 3 years.